TrackPoint Keycap Gap — Working Doc

Oct 9, 2026 · @Jordane Thomas

Handoff (as of 2026-10-10, after step 4)

Where things stand (2026-10-10): Phase 1 is complete. `build_assembly.py` (steps 1–10), `fc_helpers.py` and `sensor_sk8707.py` rebuild the whole assembly from `keycaps_test_assembly.kicad_pcb` in about two minutes on Jordane's Mac, with the switch gap, sensor hole and board size driven by the `Layout` VarSet and a 32-check report at the end (28 pass; the 4 fails are the phase 2 problems below). **Next: Phase 2, redesigning the keycaps and the sensor cap.**

Step 4 and 5 choices to carry forward: the `<REF> Params` VarSets (x, y, rot; `App::PropertyDistance` / `PropertyAngle`, since positions are negative) are created in step 4 by `make_component_varset` (doc: Document, footprint: Footprint) -> Object, which returns the existing one when called again, so step 6 reuses them and moves each into its component container. The sensor's VarSet is named `REFParams` (`*` is not allowed in object names). Footprint-local points are placed with `_on_board(params: str, x: float, y: float) -> tuple[str, str]` (rotate by rot, then offset). Each `<REF> Pads` group is built in the footprint's own frame and placed by its `<REF> Params`. `check_pcb` and `check_solder_pads` are temporary tests that step 9's `check()` replaces. In the main block, steps 6–10 are commented out and uncommented as they are filled: never stop a macro with `raise SystemExit` (tested 2026-10-10: harmless from the Macro menu, but it closes FreeCAD without saving when the macro is launched any other way).

Lesson from the 2026-10-10 failed run: FreeCAD keeps imported modules for the whole session, so an edited `sensor_sk8707.py` was ignored and step 7 failed (`unexpected keyword argument 'with_screws'`). Because the only recompute after steps 5–6 came after step 7, nothing from steps 5–6 was drawn and the raw library models showed at the origin instead. Fixes: `build_assembly.py` reloads `fc_helpers` and `sensor_sk8707` on every run, and the main block recomputes after every step.

Sandbox pre-check: FreeCAD 1.1.0 can be fetched into the sandbox from `https://github.com/FreeCAD/FreeCAD/releases/download/1.1.0/FreeCAD_1.1.0-Linux-x86_64-py311.AppImage` (820 MB; `--appimage-extract`, then `squashfs-root/usr/bin/freecadcmd script.py`). The 3D model files are not in the sandbox, so pre-checks skip the library step.

Files (all in `.../hardware/v1-mechanics/keycaps/` on the Mac, run as FreeCAD macros): `build_assembly.py` (the generator), `fc_helpers.py` (shared helpers), `sensor_sk8707.py` (sensor model), `TP_Sprintek_SK8707-01-004.kicad_mod` (footprint), plus the latest `keycaps_test_assembly.kicad_pcb`.

How we work: Claude writes one step at a time and pre-checks it headless in the sandbox (FreeCAD 1.1.0 AppImage); Jordane runs it on his Mac and pastes back the printed output. Rules: tooltips on every VarSet value, descriptions on containers, datasheet-explicit values literal, everything else relative; minimal hardcoded code; never read CAD files into chat; big files as plain attachments, not @-mentions.

Phase 2 starting points: (1) the mnhttn keycap's four clip hooks reach about 0.2 mm into Cherry's arm ends (0.006 mm^3); (2) at an equal courtyard gap of 3.05 mm the cap hits the three dactyl keycaps, KiCad's 3.61 top-bottom spacing cleared them, so the current parts need a gap between 3.05 and 3.61; (3) the dactyl keycap and the cap are 12,000-triangle mesh STEPs (slow for exact distances, `MEASURE_GAPS` is off by default). **Still open:** the write-back script (Layout gap, sensor hole and board outline into the `.kicad_pcb`); the Gerber/drill check against KiCad's own output; updating the Cherry library footprint (corrected pads, Cherry's official model, courtyard centered to ±9.5) so "Update Footprints from Library" does not undo the board; the switch model is the Tactile MX6C-T3NB, the board uses the Click K3NB.

Purpose

This is the shared memory for the TrackPoint keycap-gap project: attach it (or one tab) to a Claude session instead of re-explaining the model, the decisions and the method. Goal: minimize the gap between the four Cherry MX6C-K3NB switches around the Sprintek SK8707-01-004 FlexPoint sensor, then redesign the keycaps (mnhttn, ulp dactyl) and the sensor cap (Sprintek 109-00060) to suit.

Decisions so far

Date	Decision
2026-10-09	Pads are rebuilt from the `.kicad_pcb` footprint definitions, not sorted from the exploded KiCad STEP (cheaper, and parametric)

Date	Decision
2026-10-09	`solder_sink` = 0.02 mm (how far a component sinks into its pads to represent solder)
2026-10-09	Parameters live in FreeCAD VarSets, not spreadsheets
2026-10-09	One FCStd file; repeated bodies are in-document Links to a hidden Library group
2026-10-09	Keycap local origin = center of its switch
2026-10-09	Code is written together in small reviewed pieces; every file must be readable by a person
2026-10-09	KiCad is not installed in the sandbox (only v7 reachable; the board file is v10.99). The board file is parsed with plain Python instead
2026-10-09	FreeCAD 1.1.0 runs headless in the sandbox; your FreeCAD is the renderer
2026-10-09	The sensor cap reaches down below the board to grip the stem; the cap model is a community one (Saoto-tech, GrabCAD), not Sprintek's
2026-10-09	The SK8707-01-004 sensor is modeled from the datasheet as a parametric FreeCAD part with photo-matched colors (`sensor_sk8707.py`)
2026-10-09	`All generated code runs on Jordane's Mac in his FreeCAD; the sandbox only pre-checks scripts. (A sandbox-written FCStd with hand-made color data crashed FreeCAD on open — abandoned.)`

Date	Decision
2026-10-10	The four switches on the board now use the corrected Cherry_ULP_SMD footprint (pads and keep-out per Cherry's PCB layout drawing) and Cherry's official MX6C-T3NB model (offset 0, rotate −90 / 0 / 180). Positions, references, values, properties, nets and keycap models kept; the template switch outside the board deleted. The footprint keeps its library link Cherry:Cherry_ULP_SMD, so the library copy must be updated to match. Its large 0.3 mm courtyard is now y −10 to 9 (was ±9.5), as drawn in the new footprint.

Open questions

- ~~Keycap sources found: both have parametric sources (see Reference material). Phase 2 edits those sources instead of remaking the caps.~~
- ~~Sensor cap: it reaches below the board to attach to the stem; `cap_offset` is measured from the cap model's socket.~~
- ~~Keycap seating: measured from Cherry's own models — cavity ceiling 3.5 mm above the PCB top (see Phase 1 plan, "Keycap seating").~~
- ~~Vendor colors: harvested solids lose their STEP colors; decide whether to copy the original per-face colors or assign one color per part.~~

Reference material

Item	Source	Use
ulp dactyl keycap	jonboh/ulp-dactyl — OpenSCAD files	parametric source for phase 2
mnhttn keycap	fooker/mnhttn — Fusion `.f3d`	parametric source for phase 2
Sensor cap ("109-00060" in the model)	Saoto-tech on GrabCAD — community model, not Sprintek's	cap geometry; phase 2 base

Item	Source	Use
Cherry MX ULP (MX6C-K3NB)	frankbenoit/Cherry_MX_ULP — official Cherry documentation	stem and keycap-socket dimensions for seating
Sprintek SK8707-01	DS0048 ver. 1.04 datasheet (attached in chat)	Sensor tab

Tabs

- `Phase 1 plan` — what gets built, in what order, the hierarchy, naming and checks
- `Sensor SK8707-01-004` — datasheet dimensions and how the sensor mounts under the board
- `Methodology` — the token-economy rules, tooling, phase 2 path ranking and sources

Model facts (measured 2026-10-09)

- Original file: 1,764 objects; 22 `App::Part` containers; 43 dumb solids; 1,522 objects from an explode of the KiCad pad export, of which only 62 are solids (57 top pads, 5 bottom), 0.04 mm thick.
- Board: 47.0 × 47.6 mm, 1.6 mm; sensor footprint `REF**` at the board center = assembly origin.
- Switches SW33/34/47/48 at about (±11.0, ±11.3) mm from center; each has 7 SMD pads and 3 NPTH holes (Ø1.05 at (−5.8, 1.2), Ø1.2 at (5.8, ±3.26)). The switch body reaches 0.35 mm below the pad surface: its locating pins.
- Current gaps: LED/diode bottoms float 0.035–0.045 mm above pad tops; cap bottom sits at board level with no sensor under it.
- Keycap inner edges at 3.0–3.3 mm from center; cap radius 4.5 mm.

Ideas to circle back to

- **Press simulation with FreeCAD's Assembly workbench (Phase 2).** Keep the board and switches grounded and expression-driven; give each keycap one slider joint for its ~1.8 mm travel, then drag it through the stroke to check clearance against the sensor cap, including tilt from off-center presses. Not for Phase 1: positions come from KiCad,

joints on imported STEP faces break when models are re-imported, and a solver overwrites placement expressions. Fallback if it proves fiddly: a `keycap_press` value in Params that lowers all keycaps by expression, plus a collision check in `check()`.

- **Two versions of each keycap: a light mesh for looking, a parametric solid for designing.** The mesh-made STEPs draw every triangle edge in FreeCAD; a FreeCAD mesh (or the Links set to the Shaded display mode, which hides edges) looks clean, while the parametric source stays the design master.